Q5 NO CALCULATOR ALLOWED Q5

Answer QUESTION 5 PARTS A and B on this page.

PART A

$f''(x) = \frac{d^2y}{dx^2}$ $\quad f'(x) = (3-x)y^2$

$f''(x) = (3-x)2y + y^2(-1)$

~~$f''(1) = d^2y$~~

$f''(1) = \frac{d^2y}{dx^2}\Big|_{(1,-1)} = (3-1)2(-1) + (-1)^2(-1)$

$= (2)(-2) - 1 = -5$

PART B

~~$f'(x)$~~ ~~$f'(1) = (3-1)$~~

$\frac{dy}{dx}\Big|_{(1,-1)} = (3-1)(-1)^2 = 2.$

$f(x) + f'(x)x + \frac{f''(x)x^2}{2!}$ (ex.)

$-1 + 2(x-1) - \frac{5(x-1)^2}{2!}$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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Answer QUESTION 5 PARTS C and D on this page.

PART C

Lagrange error bound *

$$|\text{Error}| \le \frac{60}{3!}|1.1-1|^3$$

PART D

$(1,-1)$	$y+1=2(x-1)$	$(1.2,-.6)$
$(1.2,-.6)$	$y+.6=\frac{81}{125}(x-1.2)$	$(1.4,$

$$\frac{dy}{dx}=(3-x)(y^2)$$

$$(3-1.2)(.6)^2$$

$$\left(\frac{\cancel{180}^{9}}{\cancel{100}_{25}}\right)\left(\frac{\cancel{36}^{9}}{\cancel{100}_{5}}\right)=\frac{81}{125}$$

$$2\left(\frac{2}{10}\right)+\left(-1\right)$$

$$f(1.4)\approx\frac{81}{125}(1.4-1.2)-.6 \quad \frac{4}{10}-\frac{10}{10}$$

$$1.8$$

$$\left(\frac{2}{10}\right)(2)=\frac{4}{10}-\frac{10}{10}=\frac{-6}{10}$$

$$f(1.4)\approx(3-1.2)(-.6)^2(1.4-1.2)-.6$$

$$\approx(1.8)(-.6)^2(.2)-.6$$

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